

STLC:

STLC stands for:

"software testing Life Cycle."

The process of Testing an Application will termed as "STLC". The below are the stages that we are going to get as a part of STLC.

1. Requirement Analysis
2. Test plan
3. Test Case design
4. Test Environment Setup
5. Test execution
6. Test closure

1. Requirement Analysis

Based on the requirements provided by the client the team need to understand the functionalities how it's been implemented in the application along with the impacted areas.

NOTE:- As a part of test strategy document

test strategy document will be prepare and following are the different information that we are going to track as part of test strategy document.

Scope and objective:

⇒ The important of testing in current sprint are on software will be tracked under scope and objective.

Business Issues:

This mainly discuss about budget allocation between different team members.

⇒ The developers maintenance team will get 64 percentage and testing team is 36%.

Test Responsible Matrix:-

for This matrix define test to be conducted on the sprint of software

Testing scope	Yes/no	Comments
functional testing	yes	
usability testing	yes	
Compatability testing	yes	

Testing Scope	Yes/No	Comment
performance testing	Yes	
H/W Configuration testing	No	Required, but lack of resources
security testing	No	
Multilanguity	No	no requirements in current sprint/Software

Roles and Responsibilities:-

It defines the roles and responsibilities played by individual testers in the team.

Test Automation and Testing Tools:-

Here we are going to define scope of automation and tools that's we are going to use as a part of testing.

Example:- Selenium, AWS, java, ADO, etc... which are used for automation and as well as test management tool.

Defect Reporting and Defect Tracking:-

The process of tracking the reports may be through excel or by using the test management tools. Here will allocate how the team is tracking the bugs.

Test Management Tools:-

This section describes managing the testing related documents such as test cases

and Test summary reports, bug reports etc...

Test plan:-

The challenges are the risks that are faced by testing team is as below:-

- lack of time

- lack of resource

- quality gap

- delay in deliverables

- lack of documentation

- sudden changes in requirements

- lack of communication

Test plan documentation:-

Test plan document id:- It represents unique

id that may contain only digits, or name of the project

Introduction:- It describes in detail description about what the project is doing (or) sprint

test environment should be established.

Suspension Criteria:- (to interrupt from test execution)

Test environment abounded

Show stoppers

many minor bugs

Exit criteria:- (to stop test execution)

time was exceeded

all test-case testing were finished

all major closed

Staff:-

The people who are included in the testing will be listed down here.

Responsibilities:-

The responsibilities that need to be handled by respective QA will be listed down here.

Schedule:-

date and time to be followed by the testers.

Risks and assumptions:-

list of risks

approval:- need to get reviewed, and approved different stakeholders.

NOTE:- The test plan document is divided into 4 components.

1. what to test

2. How to test

3. who to test

4. when to test

Test design:-

Once the planning is done the team will start with test design based on the requirements provided by the client.

Test case design Template:-

[Text case ID] TCID

TCID | ~~Id~~² | Description | Scenario | Step No | Action |

expected result | priority (Assign to test design)

Test case design Techniques:-

The Techniques that are use in order to design the testcases. the below are the Techniques that we are going to use in order to design the test cases.

=> This are also called as "black box testing Techniques".

1. boundary value analysis

2. equivalence class partitioning technique.

3. Decision table.
4. State transition
5. Error guessing.

Boundary value Analysis:

This technique is used to test ^{any} the functional which ~~that~~ have range of the values ~~where~~ we need to consider only the boundaries instead of testing with entire input data.

Ex:- Consider any input field that will accept a range of values in between (say) 6 characters and 20 characters here in order to define the data will use

below formula $n, n-1, n+1$ do mean 6 characters so that $n=6, 6-1=5$ ⁱⁿ

$n+1=6+1=7$ valid

max 20, so that $n=20, n-1=19, n+1=21$ ^{invalid}

Equivalence class partitioning (ECP):-

This technique is used to partition (grouping) data based on input that we are use to test an functionality instead of testing with entire sets of the data.

ex:- Consider any input field that will accept a range of values in b/w 6 characters and 20 characters.

The above data will group as below

Invalid	valid	invalid
less than 6	between 6 and 20	greater than 20

Decision Table:-

This technique is used to define the test cases based on its logic and complexity and decide which test cases need to be executed. for example: consider on user login to the application once he entered valid credential home screen should be display with invalid credential error message should display

Ex:- login module

	TC1	TC2	TC3	TC4	TC5
username	valid	valid	invalid	invalid	empty
password	valid	invalid	valid	invalid	empty
home screen	y				
error		y	y	y	y

State Transition:-

This technique is used to test the software behaviours based on the states and transition between them which will occur in specific

condition.

Ex :- Consider an user try to login to an application with invalid credentials more than 3 times system will get lock. here login is an condition changing the behaviour is an state.

Error Guessing :-

Based on experience in the application during the analysis the team have capability to identify the errors that might occur during the implementation. This are about block box testing

Designing of Test cases :-

from the test case template.

Test case ID (TC-ID) :- It is an id that defines test case number which is an unique. we can use as below in order to define the test case.

Ex :- TC-001 (or) ~~use~~ ^[US] User story - 001
story - number - 001

Description:-

It defines what the functionality we are going to define as a part of the test case.

Scenario:-

one line description. what we are going to test.

Action:-

The function or Action that we are going to trigger as a part of test case.

Expected Result:-

The result that we are getting once the action is triggered.

Assigning:-

The name of the person who is going to the test case priority.

Priority:-

priority defines and what basis we need to execute the test cases

Result:-

Tracks whether test case passed or fail

TC-ID 100

TC-ID

Description

Scenario

Action

TC-001 or

US-345635-001

These test is to verify
the search result based
on the price limit
products based on price limit

Expected Result.

Action

1. Login to the application user should able to login successfully
2. Enter the product name in list of the chocolates should be search box (example: chocolate) displayed.
3. Go the price list and set $\Rightarrow$ Grid should filter the min and max values. chocolates based on the price

User story:

As an online shopper, I want to limit search results by price to select items within my budget.

This user story reflects online customer's needs specify a price range when looking for items. This feature allows customers to filter search results by price, improving the browsing experience by allowing them to exclude goods outside of their specified price range.

Bdd format

Feature: Price list

Scenario: Verify whether user is also to filter the products based on price limit.

Given launch the Flipkart application

Then Home screen should display

when user enter the chocolates and hit enter

Then chocolate should be display

And user should set the range of the price

The Grid should filter the chocolates based on the price.

②

Feature: User creation in the web application.

Scenario outline: user logs in, navigates to the new user page, and creates a new user.

- Given the user is on the login page
- when the user enters "username" <username>
- and password "<password>"
- And clicks the submit button.
- Then ^{new} user should be created

And the user should see a confirmation message. Then the user should be redirected to the home page.

when the user clicks the "New" button then the user should be redirected to the new user creation page.

when the user fills in the "required" details with first name "<first-name>", middle name "<middle-name>", and last name "<last-name>"

And clicks the submit button

Then a new user should be created

And the user should see a confirmation message

example :

username	password	first-name	middle-name	last-name
user1	pass1	John	A.	Doe
user2	pass2	Jane	B.	Smith

sample usability test cases :-

- ⇒ verify spelling and labels through the NBS
- ⇒ verify labels font style uniform next to all throughout one page of NBS
- ⇒ verify the alignment of labels and elements throughout the NBS
- ⇒ verify line spacing gap in between labels and element in throughout netbanking site
- validate mapping in between icons and corresponding tool tips in NBS site
- verify meaning of error messages in NBS site

sample testcase for crossbrowser testing :-

- verify login operation in customer expected platform
- sample testcase for hardware configuration
- verify login operation in customer expected hardware configuration

perform testcases:

verify login operation with customer expected load.

login with 1000 users

more than 1000 user

1000 users for longer duration.

Bug Life Cycle:

Whenever the actual result is not meeting the expected result that is termed as "bug".

Ex:-

Scenario:- grid is not filter based on the price limit that is set by the user.

→ login to the application

→ Enter the product name in search box (example: chocolates) and hit on enter.

→ Go to the price list and set min and max values.

Actual Result:- Grid is not filter based on the price limit.

Expected Result:- Grid should filter the chocolates based on the price list.

- The bug will go through different flows
- whenever the QA rise the bug, it will be in the meet state.
- and once the developer accept's it. there will change it to active state and once the developer rise the bug will change it into resolved state and assigned back to the resolved state and assign to QA. QA will verify the bug if every thing work.

